

# Waste2Worth

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## Abstract

Waste2Worth is a mobile application that encourages recycling through IoT-enabled smart stations and a gamified reward system. The platform allows users to submit recyclable materials, earn points, complete eco-challenges, and redeem rewards. Developed using User-Centered Design (UCD) principles, the system was evaluated through usability testing with 23 participants. Results showed positive user experience, ease of use, and increased engagement in sustainable recycling practices.

## Introduction

Waste management is becoming a major issue because of population growth and the increasing amount of waste produced every day. Traditional waste collection methods are not always effective since they depend on fixed schedules and manual work, which can increase costs and reduce recycling efficiency. Another challenge is that many people are not motivated to recycle regularly.

To help solve these problems, this project introduces Waste2Worth, a mobile platform that uses smart collection stations connected using IoT technology along with a reward system to encourage recycling. The system allows users to recycle materials, collect points, and redeem rewards, helping promote better recycling habits and greater community participation[1].

## Methodology

The Waste2Worth application was developed using a Human-Computer Interaction (HCI) approach that integrates UCD, gamification, and IoT technologies to support sustainable waste management.

The system was designed to provide an interactive, simple, and user-friendly experience that encourages users to recycle through a reward-based mechanism. The implementation followed a structured five-phase process, as illustrated in Fig. 1

### Implementation Process



Fig. 1. Waste2Worth implementation process

The system workflow, shown in Fig. 2, outlines the user journey from registration to redeeming rewards. To evaluate the system, a usability testing session was conducted covering five core tasks, with all tasks successfully completed by participants, as summarized in Table 1.

In addition, IoT-based smart collection stations equipped with weight sensors were integrated to automatically measure recyclable materials and calculate reward points in real time.

### Usability Testing - Task Results

| Task No. | Task Description         | Scenario                         | Result |
|----------|--------------------------|----------------------------------|--------|
| 1        | Account Creation         | Create new account successfully  | Passed |
| 2        | Locate Recycling Station | Locate nearby recycling stations | Passed |
| 3        | Waste Submission         | Submit recyclable materials      | Passed |
| 4        | Complete Challenges      | Complete eco-challenges          | Passed |
| 5        | Redeem Rewards           | Redeem reward points             | Passed |

Participants  
23 Total

Testing Scope  
Students, Individuals, and Organizations

Testing Method  
Heuristic Evaluation & Usability Testing

Table 1. Usability testing results and task completion scenarios.

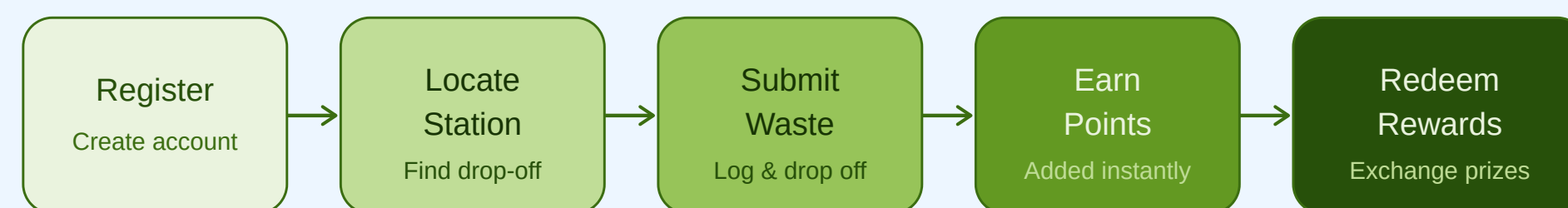


Fig. 2. Waste2Worth system workflow

## Results and Discussion

The prototype of Waste2Worth has accomplished a complete integration of all the system's core functions; a smart way to submit waste and automatically assign points based on that waste; find a location of nearby waste stations and to receive eco-challenges. These prototypes have all been developed using the guidelines of HCI and UCD [9] [15] (see fig. 3).

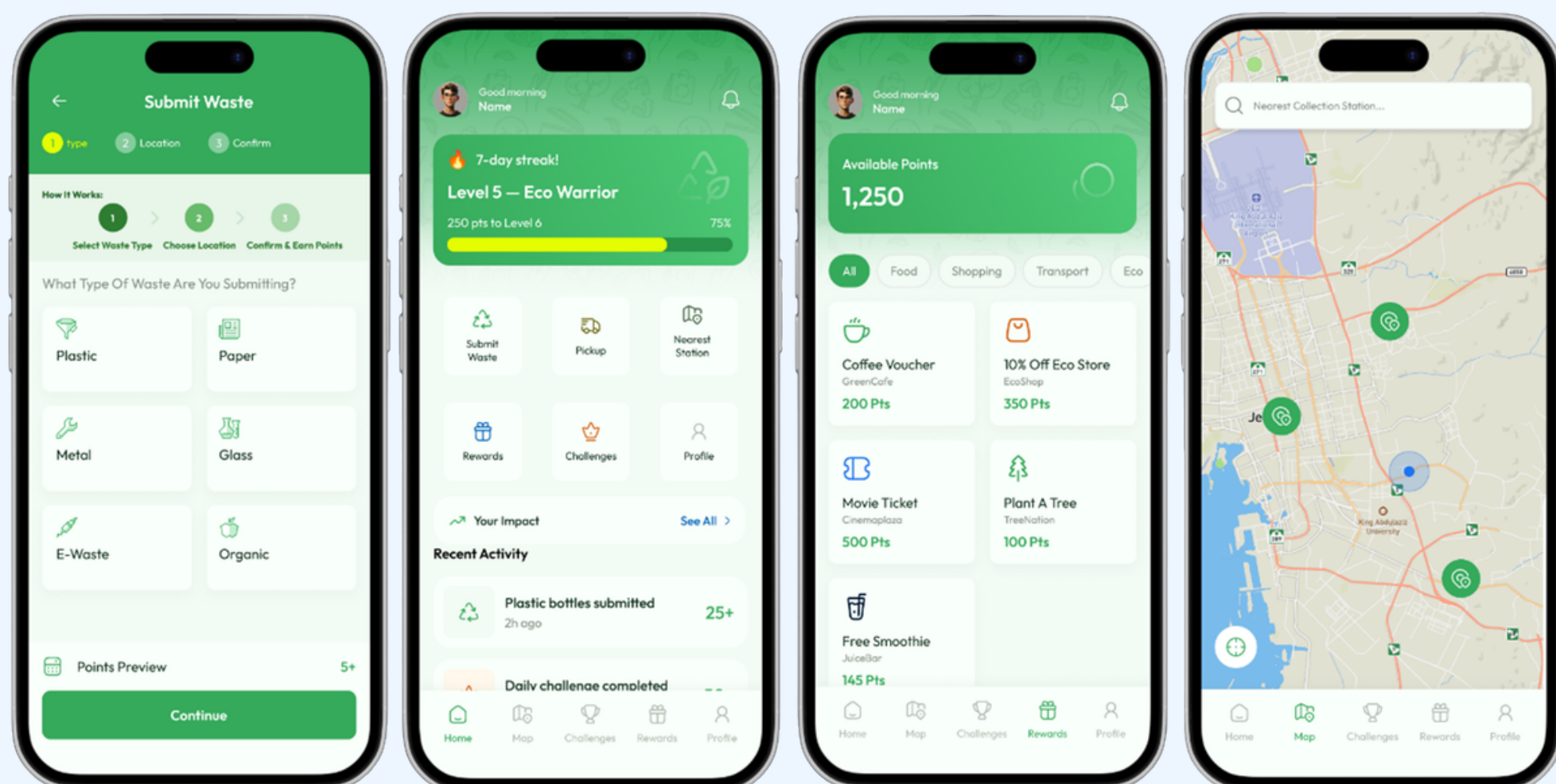


Fig. 3. Final High-Fidelity Prototype Screens of Waste2Worth.

The testing to assess the usability of the system has produced very positive results, as all participants were able to complete all designated tasks with a high level of success. In addition, by providing gamification elements (e.g. badges, streaks, challenges), motivation and engagement were enhanced. Based on results from the SUS evaluation, the usability of the Waste2Worth system has been found to be within acceptable limits, supporting the conclusion that Waste2Worth provides an easy to use and intuitive user interface, which will enable the establishment of sustainable habits for waste management.

## Conclusion

Waste2Worth has proven that by gamifying the recycling process with the use of Internet of Things (IoT) smart stations with incentive-based rewards can encourage pro-environmental behaviours amongst users. User experience assessment results validated a positive user experience, resulting in an overall positive usability and engagement with the platform.

Upcoming projects will include an Artificial Intelligence (AI) based waste classification system, blockchain reward distribution, and an inclusion of real-world applications of Waste2Worth to increase the platform's overall scalability and societal impact.

## References

